Backend · revision notes

Python

Types, collections, functions, OOP, generators, async, the GIL, tooling

1 Python basics and data types

Definition

Python is a dynamically typed, interpreted, indentation-based language. Everything is an object, and names are references to objects.

Think of Python like a kitchen with a recipe reader

1



Recipe

Source code (.py)

Plain instructions you wrote

2



Chef reading line by line

Interpreter (CPython)

Compiles to bytecode and runs it

3



Labelled jars

Variables

Names that point at objects, not boxes holding values

4



Served dish

Output

The result of the program

Type	Example	Mutable?
`int`, `float`, `bool`	`42`, `3.14`, `True`	no
`str`	`"hello"`	no
`tuple`	`(1, 2)`	no
`list`	`[1, 2, 3]`	yes
`dict`	`{"a": 1}`	yes (keys must be hashable)
`set`	`{1, 2, 3}`	yes (unique items)

Collections and comprehensions

```
nums = [1, 2, 3, 4, 5]
squares = [n * n for n in nums if n % 2 == 1]      # [1, 9, 25]
by_len = {w: len(w) for w in ["a", "bb", "ccc"]}    # dict comprehension
unique = {n % 3 for n in nums}                      # set comprehension

first, *rest = nums                                 # unpacking
print(squares, by_len, unique, first, rest)

name = "Asha"
print(f"Hello {name!r}, {len(name)} letters")       # f-string
```

- The equality operator compares values, while `is` compares identity (same object). Use `is None` for None.
- **Truthiness**: empty `[]`, `{}`, `""`, `0`, `None` are falsy.
- Strings and tuples are immutable; lists, dicts and sets are mutable.
- Slicing: `a[1:4]`, `a[::-1]` (reverse), `a[-1]` (last).

Mutable default argument

def add(x, items=[]) shares ONE list across all calls. Use `items=None` and create the list inside the function.

2 Functions, scope and decorators

Arguments, closures and a decorator

```
import functools, time

def greet(name, *args, greeting="Hi", **kwargs):
    return f"{greeting} {name} {args} {kwargs}"

def timed(fn):
    @functools.wraps(fn)                    # keep the original name and
docstring
    def wrapper(*args, **kwargs):
        start = time.perf_counter()
        result = fn(*args, **kwargs)
        print(f"{fn.__name__} took {time.perf_counter() - start:.4f}s")
        return result
    return wrapper

@timed
def slow_sum(n):
    return sum(range(n))

print(greet("Asha", 1, 2, greeting="Hello", mood="happy"))
print(slow_sum(1_000_000))
```

- `*args` collects extra positional arguments into a tuple, and the double-star parameter (`kwargs`) collects keyword arguments into a dict.
- **LEGB** scope order: Local → Enclosing → Global → Built-in. Use `nonlocal` / `global` to rebind.
- A **decorator** is a function that takes a function and returns a new one: `@timed` means `slow_sum = timed(slow_sum)`.
- Functions are first-class objects: pass them, return them, store them in dicts.
- `lambda x: x * 2` is a one-expression anonymous function.

3 Classes and OOP

Classes, dataclasses and inheritance

```python
from dataclasses import dataclass

class Animal:
    def __init__(self, name):
        self.name = name
    def speak(self):
        raise NotImplementedError
    def __repr__(self):
        return f"{type(self).__name__}({self.name!r})"

class Dog(Animal):
    def speak(self):
        return "Woof"

@dataclass
class Point:                          # __init__, __repr__, __eq__ for free
    x: float
    y: float = 0.0

print(Dog("Rex"), Dog("Rex").speak(), Point(1, 2) == Point(1, 2))
```

- `self` is the instance; `@classmethod` gets the class (`cls`); `@staticmethod` gets neither.
- **Dunder methods** customise behaviour: `__str__`, `__eq__`, `__len__`, `__iter__`, `__enter__`/`__exit__`.
- `@property` exposes a method as an attribute (computed or validated).
- Multiple inheritance resolves by the **MRO** (method resolution order, `Class.__mro__`).
- Prefer **composition** over deep inheritance; use `dataclass` for plain data holders.

4 Iterators, generators and context managers

Generators are lazy

```
def fib():
    a, b = 0, 1
    while True:
        yield a              # pauses here and remembers its state
        a, b = b, a + b

def take(n, it):
    for _, v in zip(range(n), it):
        yield v

print(list(take(10, fib())))

squares = (n * n for n in range(10 ** 9))   # generator expression: no memory
used
print(next(squares), next(squares))

from contextlib import contextmanager

@contextmanager
def opened(label):
    print("open", label)
    try:
        yield label
    finally:
        print("close", label)

with opened("db") as r:
    print("using", r)
```

- An **iterable** has `__iter__`; an **iterator** also has `__next__` and is consumed once.
- `yield` turns a function into a **generator**: values are produced on demand, so huge or infinite streams use almost no memory.
- A **context manager** (`with`) guarantees cleanup (files, locks, DB connections) even when errors happen.
- Useful tools: `enumerate`, `zip`, `itertools`, `collections.Counter`, `defaultdict`, `deque`.

5 Errors, files and modules

try / except / else / finally

```
class NotEnoughStock(Exception):
    pass

def buy(stock, qty):
    if qty > stock:
        raise NotEnoughStock(f"only {stock} left")
    return stock - qty

try:
    left = buy(3, 5)
except NotEnoughStock as e:
    print("failed:", e)
else:
    print("ok", left)                # runs only when no exception
finally:
    print("always runs")
```

- Catch specific exceptions; a bare `except:` hides bugs. Re-raise with `raise` and chain with `raise X from e`.
- Read files with `with open(path, encoding="utf-8") as f:` and iterate lines lazily.
- A **module** is a `.py` file, a **package** is a folder with modules. The `__main__` guard at the bottom of a file runs code only when the file is executed directly.
- **Virtual environments** (`python -m venv .venv`) isolate dependencies; list them in `requirements.txt` or `pyproject.toml`.
- Use **type hints** (`def f(x: int) -> str`) with mypy or pyright for safer code; they are not enforced at runtime.

6 Concurrency, the GIL and async

The GIL

CPython has a **Global Interpreter Lock**: only one thread runs Python bytecode at a time. Threads help with I/O-bound work, but CPU-bound work needs multiprocessing.

Tool	Best for	Notes
`threading`	I/O-bound (network, files)	limited by the GIL for CPU work
`multiprocessing`	CPU-bound	separate processes, real parallelism
`asyncio`	many concurrent I/O tasks	single thread, cooperative `await`

asyncio runs waits concurrently

```
import asyncio

async def fetch(name, delay):
    await asyncio.sleep(delay)          # yields control while "waiting"
    return f"{name} done"

async def main():
    results = await asyncio.gather(fetch("a", 1), fetch("b", 1), fetch("c",
1))
    print(results)                      # all three finish in about 1 second

asyncio.run(main())
```

Common mistake

Calling a blocking function (time.sleep, requests.get, a slow DB driver) inside async code freezes the whole event loop. Use async libraries or run_in_executor.

Revision

Mutable vs immutable, `is` vs equality, no mutable defaults, star-args and kwargs, decorators, generators for lazy streams, `with` for cleanup, GIL → threads for I/O and processes for CPU, asyncio for many waits, venv + type hints.

7 Rapid-fire Q&A

Q. List vs tuple?

A. Lists are mutable and for homogeneous, changing collections; tuples are immutable, hashable (if their items are) and used for fixed records and dict keys.

Q. What does `is` do versus the equality operator?

A. Equality compares values; `is` checks that two names refer to the same object. Use `is` for None.

Q. What is the mutable default argument trap?

A. Default values are evaluated once at definition time, so a default list or dict is shared between calls. Use None and create it inside.

Q. What are args and kwargs in a function signature?

A. A single-star parameter gathers extra positional arguments into a tuple; a double-star parameter gathers extra keyword arguments into a dict.

Q. What is a decorator?

A. A function that wraps another function to add behaviour; `@d` above `def f` is the same as `f = d(f)`. Use functools.wraps to keep metadata.

Q. Generator vs list?

A. A generator yields items lazily one at a time (low memory, single pass); a list builds everything in memory at once.

Q. What is the GIL?

A. A lock in CPython that lets only one thread execute Python bytecode at a time. It limits CPU-bound threading, so use multiprocessing for CPU work.

Q. When use asyncio, threads or processes?

A. asyncio for many I/O waits with async libraries, threads for blocking I/O, processes for CPU-bound work.

Q. What is a context manager?

A. An object used with `with` that sets up and always tears down a resource via __enter__/__exit__ (or @contextmanager).

Q. Shallow vs deep copy?

A. A shallow copy (`copy.copy`) copies the container but shares nested objects; a deep copy (`copy.deepcopy`) copies everything recursively.

Q. What are dunder methods?

A. Special methods like __init__, __repr__, __eq__ and __iter__ that let your objects work with built-in syntax and functions.

Q. How does Python manage memory?

A. Reference counting frees objects when the count hits zero, plus a cyclic garbage collector for reference cycles.